

Practical Insights into Data Analysis and Machine Learning

PROGRAM - 6

Develop a program to demonstrate the working of Linear Regression and Polynomial Regression. Use Boston Housing Dataset for Linear Regression and Auto MPG Dataset (for vehicle fuel efficiency prediction) for Polynomial Regression.

Objective

To implement and compare Linear Regression on the Boston Housing Dataset and Polynomial Regression on the Auto MPG Dataset for predictive modeling.

6. Introduction

Regression analysis is a fundamental technique in machine learning used to model relationships between variables and make predictions. This program demonstrates the working of Linear Regression using the Boston Housing Dataset and Polynomial Regression using the Auto MPG Dataset. The Boston Housing Dataset is used to predict house prices based on features like crime rate and number of rooms, while the Auto MPG Dataset is used to predict vehicle fuel efficiency based on attributes like horsepower and displacement. By implementing and comparing these regression techniques, we gain insights into their applicability, advantages, and limitations in different scenarios.

6.1 Linear Regression

- Linear Regression is a fundamental statistical and machine learning technique used to model the relationship between a dependent variable (target) and one or more independent variables (features).
- The goal is to find the best-fitting straight line that minimizes the difference between the predicted and actual values.
- For a single independent variable (simple linear regression):

$$y = m x + c$$

where,

y: Dependent variable (target).

x: Independent variable (feature).

m: Slope of the line (coefficient).

c: Intercept.

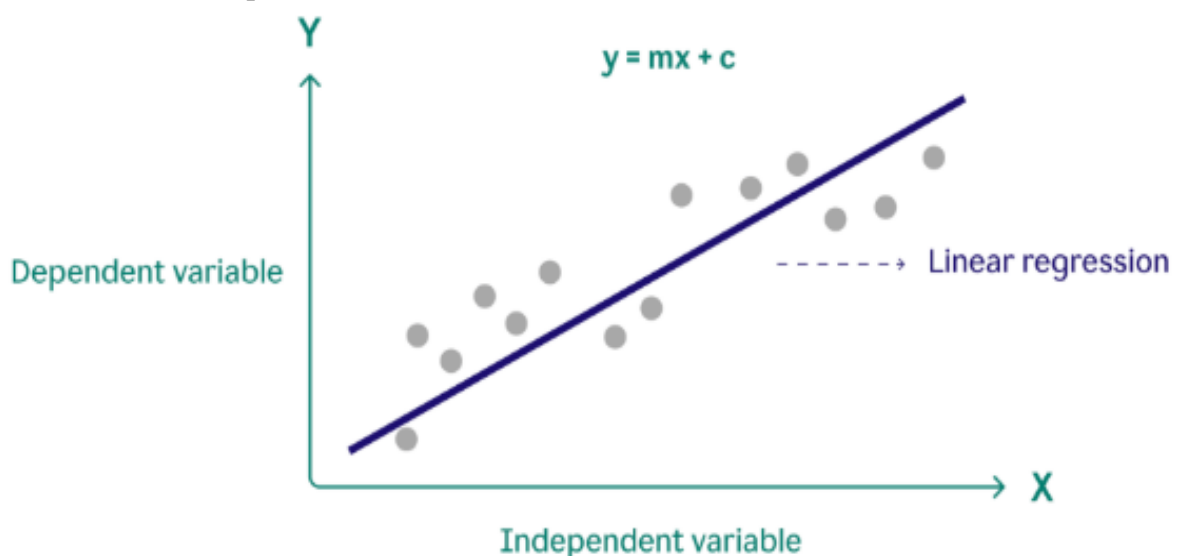


Image source: <https://www.grammarly.com/blog/ai/what-is-linear-regression/>

- For multiple independent variables (multiple linear regression)

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

where,

y : Dependent variable (target).

$x_1, x_2, \dots, x_n$: Independent variables (features).

b_0 : Intercept.

$b_1, b_2, \dots, b_n$: Coefficients for each feature.

6.2 Polynomial Regression

- Polynomial Regression is an extension of Linear Regression that models the relationship between the independent variable(s) (features) and the dependent variable as an n -th degree polynomial.
- It is used when the relationship between the variables is non-linear and cannot be adequately captured by a straight line.
- For a single independent variable, the equation is:

$$y = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$$

Where,

y : Dependent variable (target).

x : Independent variables (features).

b_0 : Intercept.

$b_1, b_2, \dots, b_n$: Coefficients for each polynomial term

n : Degree of the polynomial

6.3 Metrics for Regression Models

Mean Squared Error (MSE)

- Mean Squared Error (MSE) is one of the most commonly used metrics to evaluate the performance of a regression model.
- It measures the average squared difference between the actual values and the predicted values. MSE provides a quantitative measure of how well the model's predictions match the true data.
- MSE is calculated as the average of the squared differences between the actual (y_i) and predicted ($\hat{y}_i$) values for all data points (N):

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Where,

y_i : Actual value.

$\hat{y}_i$: Predicted value.

N : Number of data points.

- Lower MSE: Indicates better model performance, as the predicted values are closer to the actual values.
- Higher MSE: Indicates worse model performance, as the predicted values deviate more from the actual values.
- MSE = 0: Perfect model with no errors.

R-squared (R^2)

- R-squared (R^2), also known as the coefficient of determination, is a statistical measure used to evaluate the performance of a regression model.
- It quantifies how well the independent variables (features) explain the variability of the dependent variable (target).
- R-squared is a value between 0 and 1 that indicates the proportion of the variance in the dependent variable that is predictable from the independent variables.
- It is calculated as:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}$$

SS_{res} → Sum of squares of residuals (errors between actual and predicted values).

SS_{tot} → Total sum of squares (variance in the dependent variable).

- R-squared indicates how well the regression model fits the observed data.
- A higher R-squared value (closer to 1) means the model explains a large portion of the variance in the dependent variable.
- A lower R-squared value (closer to 0) means the model does not explain much of the variance.

6.4 Program

Linear Regression

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
boston_df = pd.read_csv("Boston_Housing_Data.csv")
print("Linear Regression on Boston Housing Dataset")
```

```

X = boston_df[['RM']] # Using 'RM' (average number of rooms) as the feature
y = boston_df['MEDV'] # Target variable

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Create and train the Linear Regression model
LR_model = LinearRegression()
LR_model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print(f"Mean Squared Error: {mse:.4f}")
print(f"R^2 Score: {r2:.4f}")

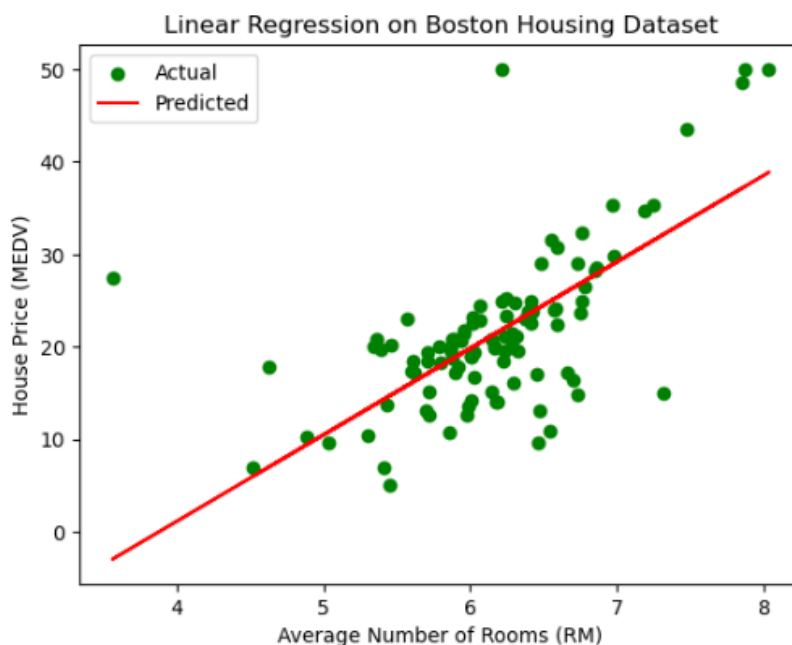
```

Mean Squared Error: 46.1448
R^2 Score: 0.3708

```

# Plot the results
plt.scatter(X_test, y_test, color='green', label='Actual')
plt.plot(X_test, y_pred, color='red', label='Predicted')
plt.xlabel('Average Number of Rooms (RM)')
plt.ylabel('House Price (MEDV)')
plt.title('Linear Regression on Boston Housing Dataset')
plt.legend()
plt.show()

```



Polynomial Regression

```
# import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load Auto MPG Dataset for Polynomial Regression
auto_df = pd.read_csv("auto-mpg.csv")
print("Polynomial Regression on Auto MPG Dataset")

# Drop rows with missing values in the Auto MPG dataset
auto_df['horsepower'] = auto_df['horsepower'].replace('?', np.nan).astype(float)
auto_df.dropna(inplace=True) #Drop rows with NaN values

X = auto_df[['horsepower']] # Using 'horsepower' as the feature
y = auto_df['mpg'] # Target variable

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Transform features to polynomial features (degree=2)
poly = PolynomialFeatures(degree=2)
X_train_poly = poly.fit_transform(X_train)
X_test_poly = poly.transform(X_test)

# Create and train the Polynomial Regression model
PR_model = LinearRegression()
PR_model.fit(X_train_poly, y_train)

# Make predictions
y_pred = PR_model.predict(X_test_poly)

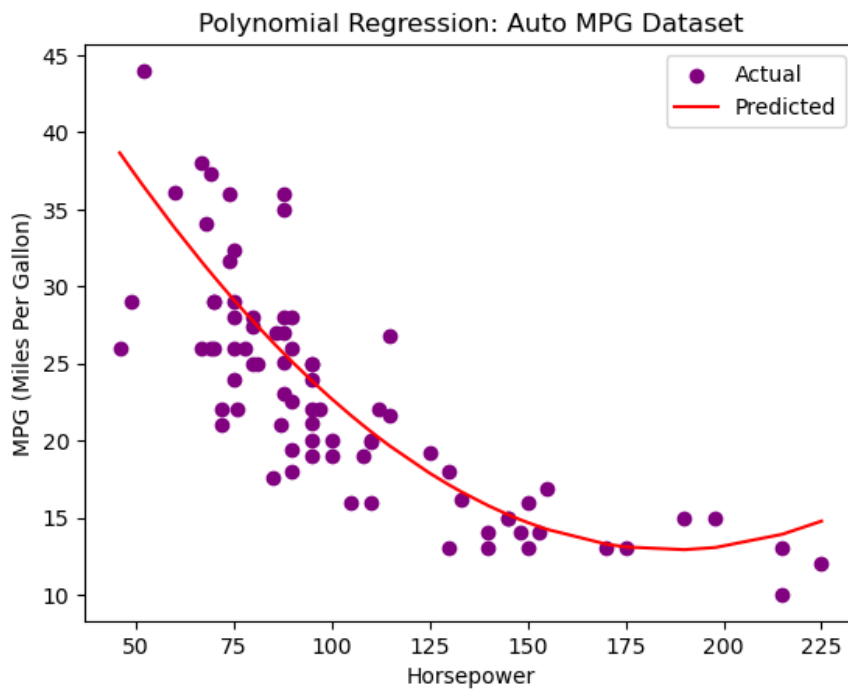
# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print(f"Mean Squared Error: {mse:.4f}")
print(f"R^2 Score: {r2:.4f}")
```

```
# Plot the results
```

```
plt.scatter(X_test, y_test, color='purple', label='Actual')
sorted_indices = X_test.squeeze().argsort()
plt.plot(X_test.iloc[sorted_indices], y_pred[sorted_indices], color='red', label='Predicted')
plt.xlabel('Horsepower')
plt.ylabel('MPG (Miles Per Gallon)')
plt.title('Polynomial Regression on Auto MPG Dataset')
plt.legend()
plt.show()
```

Mean Squared Error: 18.4170

R² Score: 0.6392



Viva Questions

- What is regression analysis, and why is it used?
- What is the difference between regression and classification?
- What are some real-world applications of regression models?
- What is the equation of a simple linear regression model?
- What do the slope (m) and intercept (c) represent in a regression equation?
- How does multiple linear regression differ from simple linear regression?
- What assumptions are made in linear regression?
- What are some limitations of linear regression?
- Why do we need polynomial features in regression models?
- What is overfitting, and how does polynomial regression lead to it?
- How can we prevent overfitting in polynomial regression?
- How do you determine the degree of the polynomial to use?
- What are the advantages and disadvantages of polynomial regression?
- What are the common evaluation metrics used for linear regression?
- How do you interpret the R^2 score in regression models?
- How does polynomial regression differ from linear regression?
- What is Mean Squared Error (MSE), and how is it calculated?
- What is R-squared (R^2), and how does it indicate model performance?
- Why is a lower MSE preferred in regression?
- Can R^2 be negative? If so, what does it indicate?